

the repo market (which allows investors to borrow short-term funds). As investors clamored for liquid assets, illiquid securities tanked and there were heavy losses for illiquidity risk strategies. Investors sought safety, so risky bonds nose-dived and credit spreads shot upward. Volatility “reached for the sky!” benefiting the purchasers of volatility protection but inflicting heavy losses on providers of it. Momentum strategy portfolios plummeted in spring 2009 when battered financial firms, shorted by momentum strategists, were buoyed by waves of bailouts and easy money.

It is precisely because factors episodically lose money in bad times that there is a long-run reward for being exposed to factor risk. *Factor premiums are rewards for investors enduring losses during bad times.*

Factors Dominate in Large Portfolios

In very large portfolios, it is very hard to find excess returns that are not related to factors. Many mispricing opportunities (or *alpha*; see chapter 10) are not scalable—we expect small pockets of inefficiency to lie in areas of the market that are illiquid or where information is not freely available. Manager decisions, both internal and external, tend to be correlated as it is hard to find truly independent portfolio strategies. Large investors hold tens of thousands of securities, and security-specific bets are swamped at the portfolio level by macro-economic and factor risks. This is not to say that large-scale security selection is impossible, but in general the bigger the portfolio, the harder it gets. A farmer, for example, may certainly be able to select a farm with the best soil and conditions for planting (farm equals security selection). But if there’s a severe drought, having chosen the best farm is not going to help (rain is the factor).

Thousands of correlated individual bets by managers effectively become large bets on factors. An equity manager going long 1,000 value-oriented stocks and underweighting 1,000 growth stocks does not have 1,000 separate bets; he has one big bet on the value-growth factor. A fixed-income manager who squeezes out the last bits of yield by finding 1,000 relatively illiquid bonds funded by short positions in 1,000 liquid bonds (through the repo market) has made one bet on an illiquidity factor.

2.5. FACTOR ATTRIBUTION

In the Professors’ Report, we showed that about 70% of all active returns on the overall fund can be explained by exposures to systematic factors. Panel A of Figure 14.3 graphs cumulated active returns for the Norwegian fund from January 1998 to September 2009. The figure shows the regime shift in 2008 that erased all of Norway’s cumulated active returns since the beginning of the sample.

Panels B and C of Figure 14.3 plot illiquidity and volatility factors, which explain a large part of the active returns of the fund. Panel B plots an *illiquidity factor*,